UNIVERSITY OF CAMBRIDGE

Data Science in Medical Imaging

MPhil in Data Intensive Science

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Coursework Minor A2: Data Science Applications to Medical Imaging

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You need to submit your answers for the three modules below (all of them).

The coursework will be submitted via a GitLab repository which will be created for you. You should write a report of up to 3000 words to accompany the software you write to solve the problems. You should place all your code and your report in this repository. The report should be in PDF format in a folder called "report". You will be provided access to the repository until the deadline above. After this you will lose access which will constitute submission of your work.

Please refer to the Handbook 5.1 for guidance on use of LLMs on the coursework, and note that you are not allowed to use them for Module 3 of this coursework.

The code associated with the coursework should be written in Python and follow best software development practice as defined by the Research Computing module. This should include:

- *Writing clear readable code that is compliant with a common style guide and uses suitable build management tools.*
- *Providing appropriate documentation that is compatible with auto-documentation tools like Doxygen/Sphinx.*
- *The project must be well structured (sensible folder structure, README.md, licence etc.) following standard best practice.*

- *Appropriate and robust unit tests for automatic validation.*
- *Uses appropriate version control best practice, including branching for development and testing, and commit hooks to protect "main".*
- *Appropriate containerisation to ensure portability of the project to other computers and operating systems if necessary.*

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MODULE 1: Tomographic Reconstruction

Exercise 1.1: Reducing Dose [5]

Reducing the dose of CT image acquisition is an active field of research. There are two clear ways to reduce the dose delivered: reducing the source intensity and reducing the total number of projections.

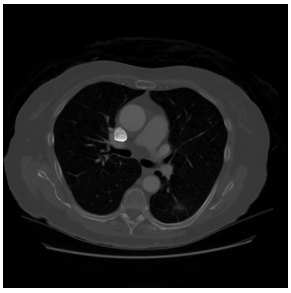

Figure 1: CT image to use

b Simulate Gaussian ($\mu = 0$, $\sigma = 0.05$) and Poisson noise at three different levels ($I_0 \in [10^5, 10^3, 10^2]$). Simulate these sinograms at different total number of angles, 360, 90, 20.

c Reconstruct all sinograms, first with FBP, and then Gradient Descent. What is the performance? When do they do better or worse? Display reconstruction error and image quality metrics and do a comparative analysis. What does each scanning procedure and algorithm mean in clinical practice for the patients? what are their impact?

d What could we do to get better images? Give 2 or 3 examples of methods.

Exercise 1.2: Limited Angle tomography [5]

Sometimes due to measurement constraints (e.g. a robot arm in image guided surgery) or for scanner design (e.g. Digital Breast Tomosynthesis).

a Simulate noisy sinograms at three different angular ranges, 180, 120, 40.

b Repeat the simulations experiment from 1.1, but this time for this limited angle range. What differences do you observe in the limited angle case?

c What is the mathematical principle that describes this effect? What describes if you can see an object edge in CT?

Exercise 1.3: Reconstruction

a Filtered back-projection (FBP) uses a high-pass (Ram-lak) filter. There are other filters in the literature. Which ones? [1]

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b Code and two alternative filters and compare them to the standard one in 1 of the low dose examples of exercise 1.1. [4]

c Gradient descend updates the image using all projections in each iteration. Instead, one can split the projections in "mini-batches" and update the image several times in each iteration (one per batch). Often this is called OS-SART in CT. Code this and compare it to gradient descend (aka SIRT) for one of the low dose experiments. [4]

d Why is MLEM or OSEM the algorithm of choice in PET, yet CT algorithms use Gradient descend (aka SIRT) like algorithms? [1]

MODULE 2: MRI Image Denoising

For this module you will work with k-space data (Fourier space data) from MRI – and deal with a few common artifacts. The data contains a 3D knee image with six coil receivers.

Exercise 2.1 Visualisation and identifying noise

- Load the complex data "kspace.npy" using the `np.load()` routine. Determine which array dimension is the coil dimension. [1]
- Create an image of the magnitude of k-space for each coil. You can use `np.log1p()` to scale the kspace data for easier visualisation. [1]
- Transpose the data into image space using the Fourier transform. Create and show a magnitude and phase image from one coil. [1]
- Show the magnitude images from all coils. [1]
- Combine the data from all of the coils into a single image (in image space). Perform a square of the image from each coil, sum the magnitudes, and the calculate the square root of the final result. Show this image. [2]
- Write about your observations. [2]

Exercise 2.2 Removing noise

- Choose three denoising methods using the image space data, and show the effects of the denoising for all coils. Denoising methods can include mean, bilateral, wavelet, or any other filter. A number of filters can be found in scikit-image. "gaussian_filter" can be found in scipy.ndimage. Comment on the different denoising methods. [10]

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- Start from the original noisy k-space. For the first coil, apply a low-pass Butterworth filter within k-space and show the new image, showing both magnitude and phase images. How does this denoising method compare to the image-based methods? [6]
- Using one of denoising methods on the images, and create a denoised image from the combined coil images. [4]
- Compare and discuss on the different denoising methods. Discuss any methods that might be able to further improve the combined image quality. [2]

The Butterworth filter can be defined as below. Make sure your k-space has a bright central intensity.:

```
def butterworth_lowpass_filter(shape, D0=30, n=2):
    P, Q = shape[0], shape[1]
    u = np.arange(P) - P // 2
    v = np.arange(Q) - Q // 2
    U, V = np.meshgrid(u, v, indexing='ij')
    D = np.sqrt(U**2 + V**2)
    H = 1 / (1 + (D / D0) ** (2 * n))
    return H
```

MODULE 3: Image segmentation methods.

Exercise 3.1: Critical comparison of methods (30 marks).

a Find 2 recent (not older than 7 years) examples of academic publications published in high-impact journals that describe segmentation tasks on medical images. It can be segmentations of different things (e.g. bones, tumours, etc.) and in any imaging modality (e.g. CT, MRI, etc.). Select one example of such publications in which processing-based algorithms (non ML/DL methods) are used and one examples of such publications in which data-driven (ML and/or DL) algorithms are used. You are not allowed to select any publications referenced or used as examples within the lectures. [0]

b For the two publications selected in part a), answer the following:

- What was done in each paper? Write a summary of a maximum of 10 lines for each paper. [5]
- Which methods work better/worse in different scenarios? Critically compare the methods studied, highlighting advantages and disadvantages, strengths and limitations. Write a maximum of one page. [5]

c What would you do to improve one of them? Propose a methodology that builds on those analysed and aims to resolve identified issues or to improve performance.

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It can be a different method, a combination of methods, or a workflow for further testing and optimising. Focus only on improving a single study/paper. Write a maximum of half a page. [10]

You should not use Generative AI to complete this exercise. Suspicion of its usage will result in losing marks. Text beyond the limits specified will not be considered (images, if used, will not be counted against the limit).

END OF PAPER

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